ENGINEERING SPECIFICATION • REF: PXC-DSA-IFM-001

System Requirements Specification (SRS)

Daily Statement Application • Multi-Branch, Offline-Capable PWA with Background Web Push

CLIENT	ENGINEERING FIRM	ARCHITECTURE TARGET	DOCUMENT VERSION
Indo Fish Meal & Oil Co. Pvt. Ltd.	**Pixelcodes**	**PWA + Node.js + PostgreSQL**	**v1.2 (Production Baseline)**

1. Executive Summary & Scope Overview

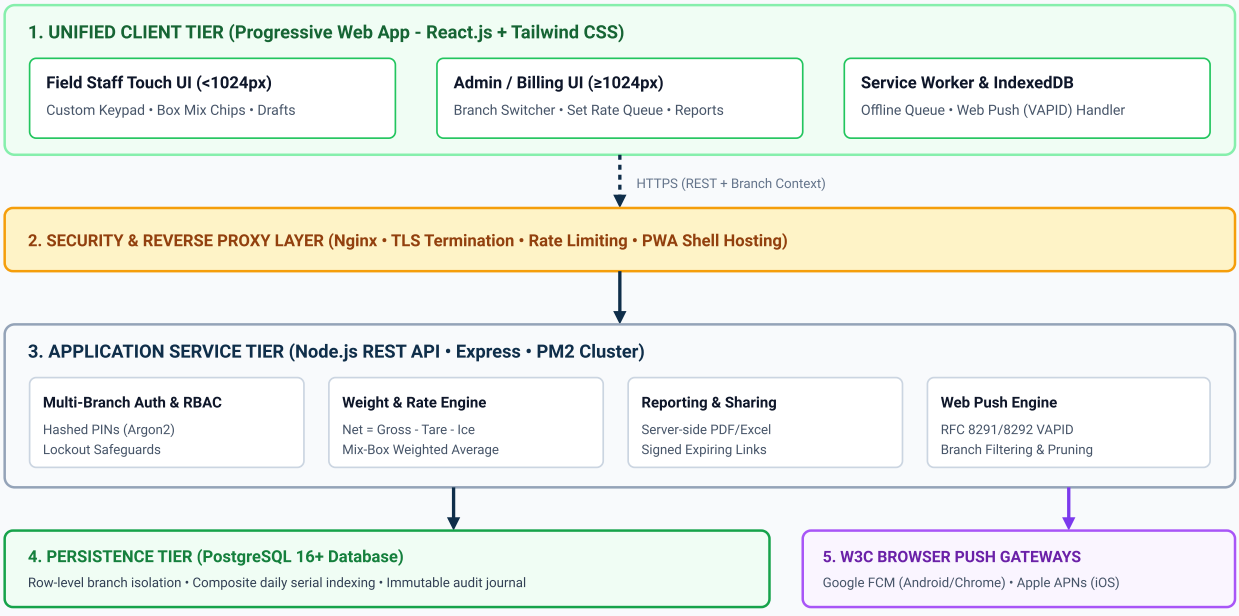
This document details the engineering specifications and architectural requirements for the **Daily Statement Application**. The application digitizes raw fish procurement, vehicular gross/tare/ice weight capture at industrial weighbridges, administrative rate fixing, billing rate adjustments, multi-branch daily statement reporting, and automated client summary sharing via WhatsApp.

The solution is delivered as an end-to-end **Progressive Web App (PWA)** serving both mobile touch interfaces for field weighbridge operators and wide-screen sidebar management dashboards for administrative personnel from a single unified codebase. The architecture is engineered around **multi-branch tenancy**, **offline-first local persistence**, and **W3C Web Push notifications** utilizing standard operating-system gateways.

2. High-Level System Architecture

The system follows a three-tier decoupled architecture: a unified client layer, a reverse-proxy security gateway, an asynchronous Node.js application server, and a relational persistence layer. All requests carry branch context to enforce rigorous tenant isolation.

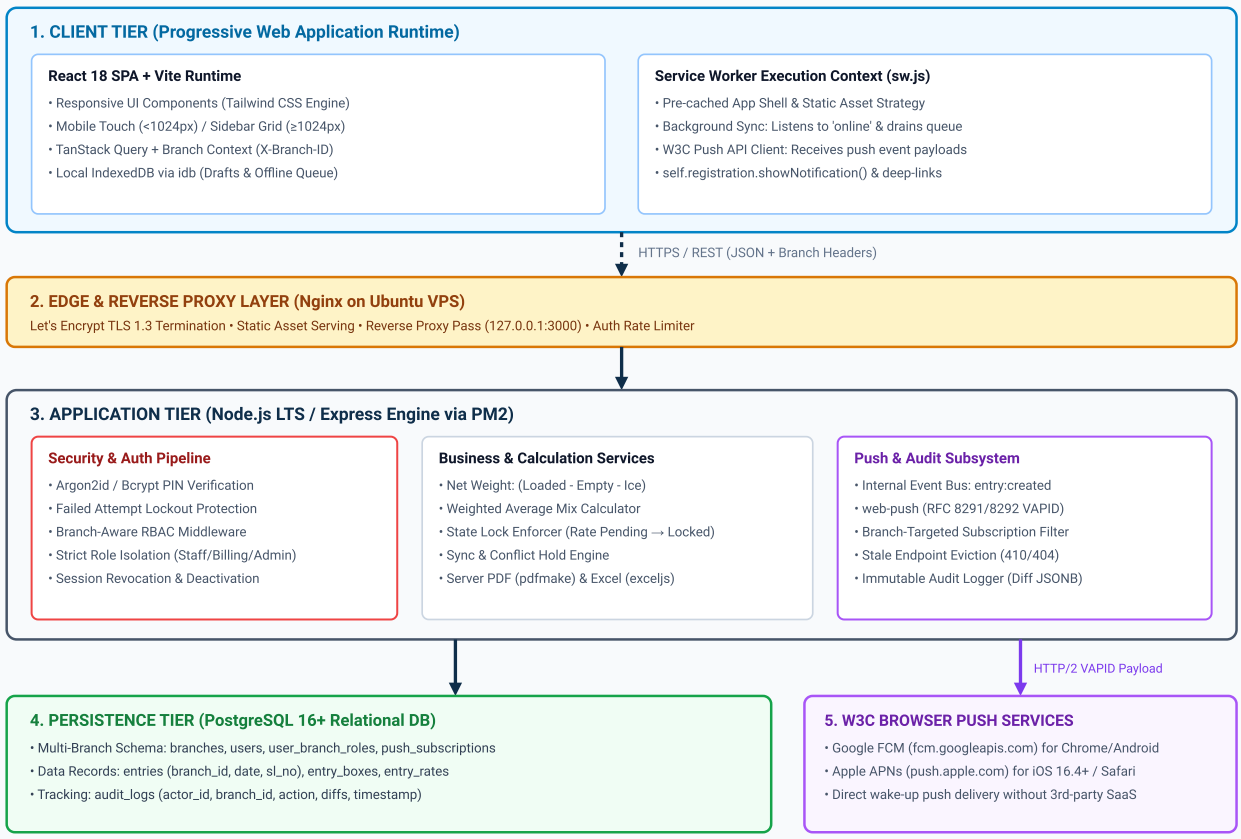
FIGURE 2.1: UNIFIED END-TO-END SYSTEM ARCHITECTURE



3. Detailed Technical Architecture & Component Pipelines

To provide clear separation of concerns across runtime contexts, the diagram below details the concrete execution boundaries between the single-page application (SPA), the service worker background thread, the Nginx reverse proxy, the Node.js application services, the persistence tier, and upstream W3C push servers.

FIGURE 3.1: TECHNICAL ARCHITECTURE & COMPONENT EXECUTION PIPELINES



4. Multi-Branch & Role-Based Access Control (RBAC)

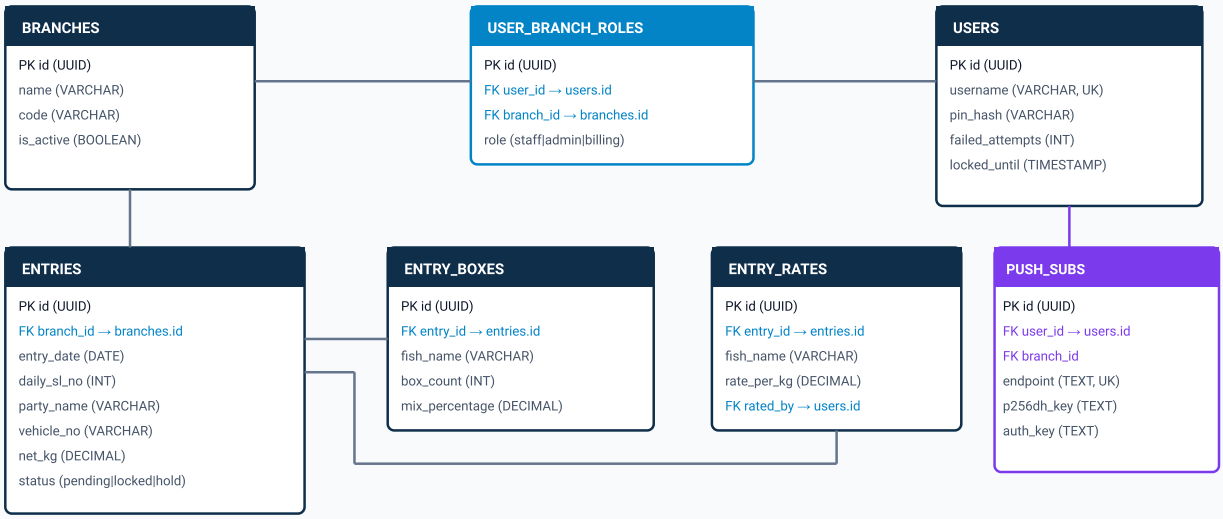
The system allows a single enterprise entity to operate multiple physical branches (e.g., Ullal Unit, Mangaluru Port). User permissions are bound to branch contexts via an association table.

Role	Branch Association	Permissions & Capabilities	Operational Boundary
Staff	Single Branch or Multi-Branch	Record daily purchase entries, vehicle numbers, loaded/empty/ice weights. Edit entries until rate is assigned.	Locked from viewing rates, viewing day statements, or accessing accounts.
Admin	All Branches or Designated Branch	Receive new entry push alerts, set per-fish purchase rates, review statements, export PDF/Excel, manage users.	Can unlock records, manage master lists, and resolve late-sync conflicts.
Billing	Single or Multi-Branch	Inspect day statements once rates are approved; adjust fish rates; export reports; share via WhatsApp.	Cannot create raw entries, weigh vehicles, or access user accounts.

4. Database Architecture & Entity Relationships

The persistence layer utilizes PostgreSQL with composite indexes on `(branch_id, entry_date, daily_sl_no)` to guarantee isolated numbering sequences per branch.

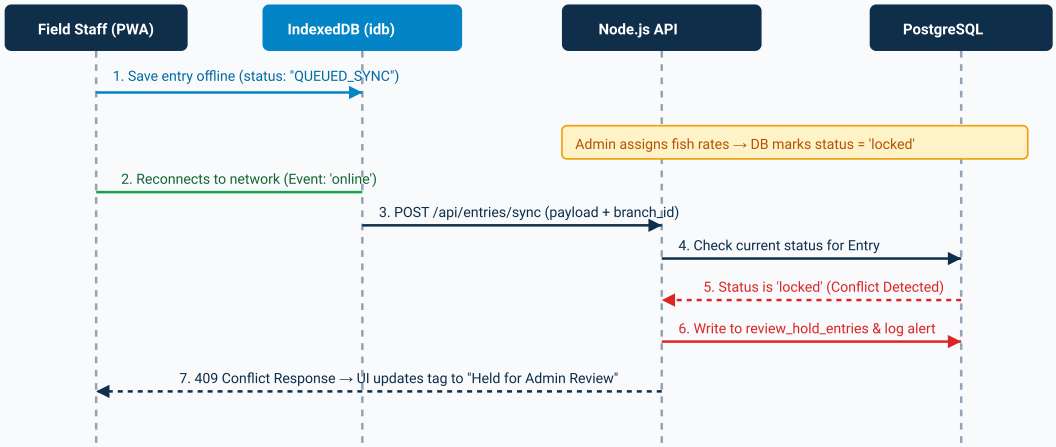
FIGURE 4.1: RELATIONAL ENTITY-RELATIONSHIP DIAGRAM



5. Offline Synchronization & Conflict Lifecycle

Weighbridges frequently suffer intermittent cellular connectivity. The PWA operates an offline-first storage engine using browser IndexedDB. When connectivity is restored, items synchronize automatically. If an entry has been rated by an Admin prior to a late offline edit, the edit is diverted to an **Admin Review Hold** queue to prevent silent overwrites.

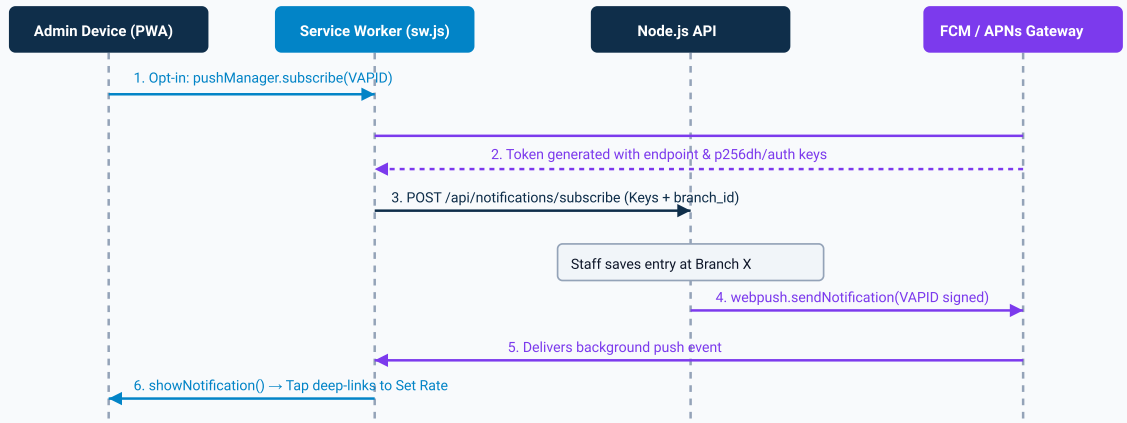
FIGURE 5.1: OFFLINE SYNC & RATE-LOCK CONFLICT SEQUENCE



6. Native W3C Web Push Notification Subsystem

To eliminate expensive notification gateway subscriptions, the system implements standard RFC 8291/8292 Web Push. Push messages are encrypted and dispatched directly to official platform endpoints (Google FCM and Apple APNs), waking background or closed devices.

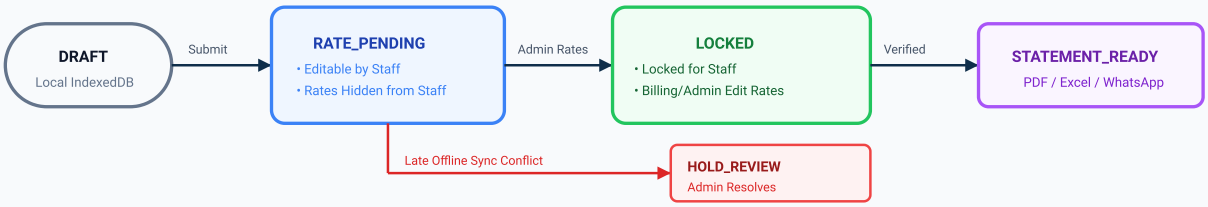
FIGURE 6.1: NATIVE W3C WEB PUSH NOTIFICATION PROTOCOL FLOW



7. Entry State Machine & Workflow Logic

Every purchase transaction transitions through strictly enforced lifecycle states. Staff editing privileges terminate the moment an Admin sets purchase rates.

FIGURE 7.1: PURCHASE ENTRY STATE MACHINE TRANSITION MODEL



8. Functional Requirements Specification

FR-01: Authentication & Access Safeguards

- Users authenticate via an auto-issued username and 4-digit numeric PIN using an on-screen keypad.
- PINs are irreversibly hashed server-side using `Argon2id` or `bcrypt` with strong work factors.
- Failed attempts are strictly bounded: 5 successive invalid entries result in a 15-minute account lockout.
- Admin can remotely reset PINs, revoke active sessions, and activate or deactivate accounts on demand.

FR-02: Purchase Entry & Weighbridge Computation

- Operators record vendor party (with search-as-you-type and quick-pick chips), fish types, box counts, and optional C-values.
- Boxes allow multi-fish mixing by percentage; client and server validation verify that the percentage sum equals exactly 100%.
- Weighbridge entry captures gross loaded kg, empty tare kg, and water/ice deduction kg:
  `Net Weight = Loaded kg - Empty kg - Ice/Water kg`
- Live preview generates standard narrative remarks (e.g., *"50 BX CHAMBAN + 20 BX (80% SARDINE, 20% MIX)"*).

FR-03: Percentage-Weighted Rate Calculation

- When an entry contains mixed-fish boxes, the system derives the weighted average rate:
  `Average Rate = ∑(Box Count × Fish % × Fish Rate) / ∑(Box Count × Fish %)`
- Recalculates dynamically whenever an Admin or Billing user updates individual fish rates.
- Staff access tokens cannot query or inspect calculated or individual rates.

FR-04: Statement Filtering, Export & WhatsApp Sharing

- Statement views provide filtering by date range (Today, Yesterday, Last 7 Days, Custom), party, branch, and status.
- Exports produce formatted PDF statements (using clean print formatting) and Excel files (using `exceljs`).
- WhatsApp sharing opens pre-filled web or native chat links addressed to the vendor's saved phone number, containing a secure, time-expiring (7-day default) link to that specific party's daily summary.

FR-05: Comprehensive Audit Journal

- The system persists an append-only audit trail recording every change to entries, weights, rates, and branch configurations.
- Logs capture the user ID, branch context, timestamp, action category, old state snapshot, and new state snapshot.

9. Non-Functional & Technical Stack Baseline

Layer / Dimension	Technology & Tooling	Specification Target
Frontend Engine	React.js (Vite), Tailwind CSS, TanStack Query	100% PWA-compliant; installable on Android, iOS, Windows, macOS.
Local Storage	IndexedDB via `idb` Promise wrapper	Offline queue holds up to 500 entries locally with automated sync.
Backend API	Node.js LTS (v20+), Express.js, PM2 Cluster	Stateless REST API; sub-100ms response time on local endpoints.
Database Tier	PostgreSQL 16+ on Ubuntu 24.04 LTS VPS	ACID compliance; nightly compressed database backups offsite.
Web Push Engine	Node `web-push` (RFC 8291/8292 VAPID)	Zero ongoing subscriptions; sends directly to FCM and APNs.
Hosting & Edge	DigitalOcean VPS, Nginx, Let's Encrypt TLS	A+ SSL rating, strict UFW firewall, ports 80/443 open only.

> **Implementation Sign-off Notice:** This specification serves as the formal baseline for Indo Fish Meal & Oil Co. Pvt. Ltd. and Pixelcodes engineering teams. System development, schema migrations, and test case coverage shall adhere strictly to the interfaces and state models defined herein.